

Anish Rao Toorpu

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EDUCATION

North Carolina State University, Raleigh NC, USA

August 2023 - May 2025

Master of **Computer Science**

GPA: 4.0/4.0

Coursework: Neural Networks and Deep Learning, Design and Analysis of Algorithms, Software Engineering, Automated Learning and Data Analysis, Object-Oriented Design & Development, User Experience, Operating systems.

Anurag University, Hyderabad, India

August 2019 - May 2023

Bachelor of Technology in **Artificial Intelligence**

GPA: 8.66/10.0

TECHNICAL SKILLS

Languages: Python, C#, C++, Java, TypeScript, SQL, Go

Web Technologies: HTML, CSS, React, Node.js, REST APIs, GraphQL, AWS, GCP

Databases: MySQL, Oracle, Redis, PostgreSQL, MongoDB, Firebase

Tools/ Frameworks: Git, Ruby on Rails, Spring Boot, Kafka, Kubernetes, Jira, Microservices

Others: R-Studio, Linux, Radius, Unix, Docker, CI/CD, Postman, DevOps

EXPERIENCE

Software Engineer, iSimcha, LLC

May 2025 - Present

- Built distributed LLM inference pipelines (React TSX + Python + Docker on GCP), improving throughput by 30% and enabling scalable deployments.
- Fine-tuned domain-specific LLM's on healthcare datasets reducing model perplexity by 18% and improving translation accuracy for complex medical terminology and integrated LiteLLM for inference calls.
- Deployed Google Cloud Functions on Firestore to orchestrate model workflows and store results in Firebase, and collaborated on evaluation protocols to improve model selection efficiency by 25%.

Software Engineer, North Carolina State University

September 2023 - May 2025

- Engineered and implemented an LSTM-based Anomaly Detection System for the BTEC Water Treatment Facility, elevating operational efficiency by **40%** through accurate forecasting and issue detection.
- Refined secure token retrieval and enhanced GraphQL request handling, enhancing data security and reducing data retrieval issues by **30%**.
- Designed visualization tools and automatic email notifications, facilitating operational management and timely responses to anomalies, resulting in a **25%** improvement in system reliability.
- Developed Python-based reporting backend on AWS (Flask, S3, RDS) to automate student mental-health and transfer-readiness analytics, reducing manual effort and improving accuracy.
- Built serverless APIs with AWS Lambda and API Gateway, secured with IAM roles, and deployed via CI/CD pipelines using CodePipeline, Docker, and CloudFormation for scalable, reliable delivery.

Software Engineer, Cognizant Technology Solutions

February 2023 - June 2023

- Architected and deployed scalable loan management systems using Java and Spring Boot, driving a **30%** increase in operational efficiency.
- Collaborated with cross-functional teams using Git and Jenkins to identify and resolve bugs in the loan management applications, resulting in a 20% reduction in customer complaints and improved system stability.
- Automated critical processes with Kafka for real-time event streaming, minimizing manual efforts and boosting productivity while achieving notable cost reductions.

PROJECTS

Event Management System (Ruby on Rails)

- Developed a comprehensive event management system using Ruby on Rails, allowing users to manage events, book tickets and leave reviews.
- Executed functionalities for both admin and attendees, including event creation, ticket booking, attendee management, and review systems, resulting in a **25%** increase in user engagement and satisfaction.
- Integrated bonus features such as filtering attendees by event name and allowing users to book tickets for other users.

Data Query System (Spring Boot, React, MySQL, OAuth 2.0 & JWT)

- Spearheaded the development of a streamlined data query system, facilitating efficient access to student information from existing event records.
- Implemented a robust solution utilizing Spring Boot for backend operations and React for frontend interactions, seamlessly integrating a MySQL database for efficient management and analysis of student and event data.
- Integrated OAuth 2.0 and JWT (JSON Web Tokens) for secure user authentication and authorization, enabling secure access across multiple platforms while protecting sensitive data.